to the Elixir compiler. Plus, it can have problems with accuracy and speed. It's recommended that you use Dialyzer, but it's not a true types support either.

Dialyzer uses a concept called Typespecs[12] to define types—in addition to what it can automatically determine from your code. You may see `@spec` definitions in Elixir codebases that you work on. They look like this:

```
# Example from Typespecs docs
@spec long_word?(word()) :: boolean()
def long_word?(word) when is_binary(word) do
  String.length(word) > 8
end
```

Typespecs are used for documentation and by tools like Dialyzer, but there may be more uses for them in the future.

José Valim opened the discussion for a real type system in Elixir with a blog post[13] and presentation at ElixirConf 2022. The team has enlisted the help of PhD researchers to develop an Elixir type system based on set-theoretic types. This is big news because it was largely considered impossible to add true types support to Erlang and Elixir.

There's no guarantee that this research will result in a types implementation. It will only be added if it's performant, accurate, and easy to use. As of January 2024, José tweeted[14] that "Elixir is, officially, a gradually typed language." So the future is bright, and the community is hoping to see a true types system in Elixir one day!

Wrapping Up

Your Elixir toolbox has quite a few tools in it, but it's nowhere near complete yet. There's so much to learn, and there are different paths that you can take based on your interests. This is a good thing though because it means you can expand your knowledge and interests over time.

The first thing you'll need to do once you write your first Elixir app is deploy it somewhere. Use Mix releases to create your release and package it up into a container. You'll be able to easily deploy to platforms like AWS, GCP, Fly.io, or Render. Once your application is deployed, make sure that you have proper observability in place, such as application logging, performance monitoring, and bug tracking.

12. https://hexdocs.pm/elixir/typespecs.html
13. https://elixir-lang.org/blog/2022/10/05/my-future-with-elixir-set-theoretic-types/
14. https://twitter.com/josevalim/status/1744395345872683471